Vishnu Shreeram M. P. - Research Summary

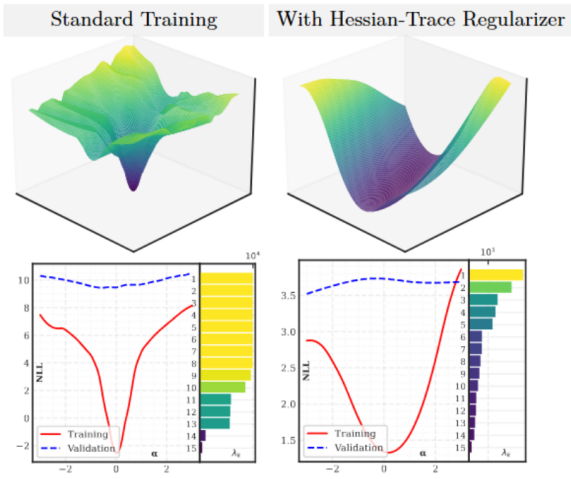
Project: Tractable Sharpness-Aware Learning of Probabilistic Circuits
(Accepted for Oral Presentation, AAAI 2026, Singapore)

1. The Problem & Motivation

Modern neural networks often achieve impressive predictive performance, yet they tend to be highly confident even when they are wrong. While probabilistic modelling explicitly represents uncertainty, traditional models provide principled estimates but are often difficult to scale. To address this, we focused on Probabilistic Circuits (PCs), an alternative architecture designed for tractable, exact inference.

2. Core Methodology

Our research focused on making the tractable learning of PCs explicitly aware of geometric sharpness, which helps prevent overfitting when data is limited. We achieved this by leveraging the structural properties of Probabilistic Circuits, which allow the exact computation of the Hessian trace in linear time using only first-order edge flows obtained from a single forward-backward pass. Because the absolute trace of the Hessian mathematically simplifies down to the sum of the squared partial derivatives, and these circuit flows are already calculated during standard training via back-propagation, this critical second-order curvature information is essentially acquired for free. This allows us to efficiently guide the models toward flatter, more robust optima.



3. My Technical Contributions

While the broader team formulated the theoretical foundations, my primary focus was translating these abstract models into deployable code and conducting disciplined experimentation.

- **Algorithm Implementation:** I implemented the theoretical formulations into major PC frameworks, specifically Einsum Networks and Pyjuice. Using PyTorch, I modified the internal loss functions, integrated our custom regularizer, and propagated these architectural modifications throughout the libraries to ensure correct gradient flow and structural integrity.
- **Empirical Validation:** I designed and executed systematic ablation studies across multiple datasets to benchmark our hybrid method. I proved empirically that our approach outperforms standard Expectation-Maximization (EM) and Stochastic Gradient Descent (SGD) baselines in data-scarce regimes.
- **Systems & Compute Optimization:** I deployed and managed these large-scale experiments across IIT Palakkad's Madhava HPC GPU cluster. Through this rigorous testing, I demonstrated that our method scales linearly with model depth, contrasting sharply with the exponential memory blow-up typical of standard automatic differentiation.

Exploring how structured probabilistic reasoning and generative modelling can combine to build more reliable systems deeply interests me. My work on Probabilistic Circuits required fundamentally questioning standard architectural biases to achieve tractable inference and circumvent the exponential memory costs of standard autodiff. I am eager to apply this exact efficiency-first mindset to dense retrieval at MSR. I want to analyze where deep transformer encoders hit low-latency bottlenecks in production and explore scalable, alternative architectures that preserve deep-encoder accuracy without breaking the computational budget.